

Public Economics (ECON 131)

Section #8: Externalities

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you cannot reallocate resources to improve one individual's situation w/o harming someone else

private market equilibrium is pareto efficient

market failure occurs when one of the assumptions of the first welfare theorem fails

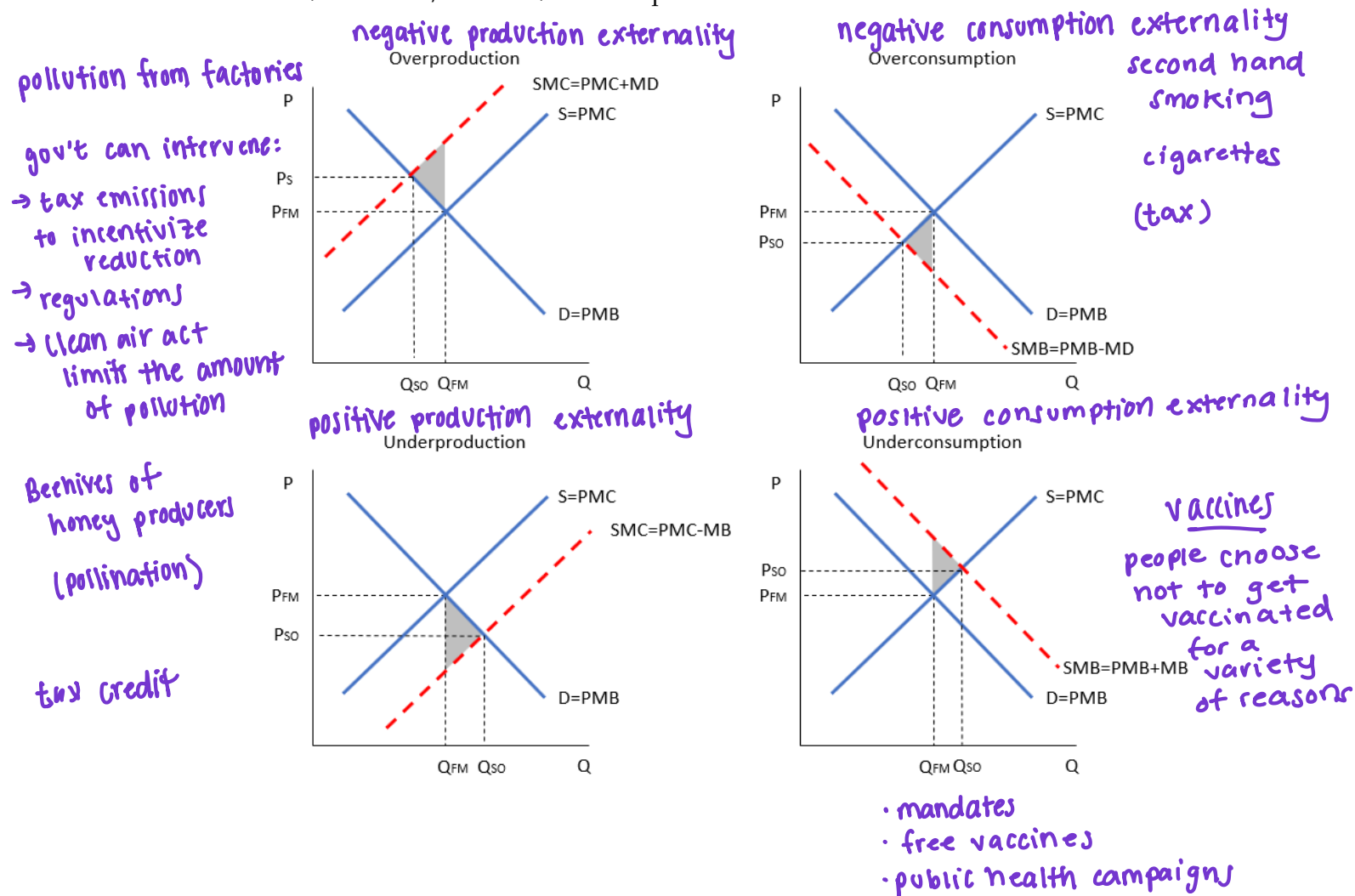
one question we ask in public economics:
when should the gov't intervene in the economy
• redistribution • market failures

1 Externalities

1.1 Key concepts

- **Market failure:** A problem that causes the market economy to deliver an outcome that does not maximize efficiency. → **DWL!**
→ market economy sometimes fails to deliver efficient outcomes in these cases, gov't intervention can help!
 - Externalities, asymmetric information, individual failures
- **Externality:** An externality arises when the action of an agent directly makes other agents better off (positive externality) or worse off (negative externality), with no costs or benefits to the acting agent. ↳ people are not always rational (do not save for retirement)
a steel plant that pollutes a river (neg externality)
- Externalities can occur because of production or consumption
- Private vs. social marginal cost:
 - Private marginal cost (PMC): is the direct marginal cost of producing/consuming an additional unit of a good
 - Social marginal cost (SMC): is the PMC + marginal damages (MD), ie the PMC + any additional costs associated with production/consumption that are imposed on others in the economy

- In a free market, equilibrium where $PMB = PMC$
- Social optimum at $SMB = SMC$
- If $PMB \neq SMB$ and/or $PMC \neq SMC \Rightarrow$ private market leads to inefficient outcome



- We say an agent **internalizes the externality** when, due either to private negotiations or government actions, the price of an action fully reflects the external costs or benefits of that action.
 - Private negotiations can bring result in socially optimal quantity as long as (a) there are well-defined property rights (b) costless bargaining, and (c) competitive markets (**Coase theorem**—but see lecture slides for some limitations)
 - Public policy can enter when Coase conditions are not met:
 - * price policy (tax or subsidy)
 - * quantity regulation (command and control or tradable permits)

1.2 Practice Problems

1.2.1 Based on Gruber Ch.05, Q.13

Suppose that demand for a product is $Q = 1,000 - 4P$ and supply is $Q = -200 + 2P$. Furthermore, suppose that the marginal external damage of this product is \$6 per unit.

$MD = \$6$ per unit

(a) How many more units of this product will the free market produce than is socially optimal?

Step I: Find the P and Q in the free market

Demand = Supply

$$Q_D = 1,000 - 4P = -200 + 2P = Q_S$$

$$1,200 = 6P \Rightarrow P_{FM} = 200$$

$$Q_D = Q_S = -200 + 2(200) = 200$$

here, I show two different mathematical methods. you can use either one.

marginal damage
so, this is a negative externality!

now, is it overproduction or overconsumption? we don't know! will show both

Step II:

{tax on consumers}

② externality price added to producers (overproduction)

① externality price added to consumers (overconsumption)

$$P^D = P^S + 6 \text{ (price consumers pay)}$$

price producers receive

$$Q^D = 1,000 - 4(P^S + 6) = -200 + 2P^S = Q^S$$

$$\Rightarrow 1,200 - 4P^S - 24 = 2P^S$$

$$\Rightarrow 1,176 = 6P^S$$

$$P_{so} = 196$$

$$Q_{so} = -200 + 2(196)$$

$$= -200 + 392$$

$$= 192$$

here, consumers internalize the externality

both solutions give us that

$$Q_{so} = 192$$

$$Q_{FM} - Q_{so} = 200 - 192 = 8$$

free market will produce 8 units more than is socially optimal

rewrite to get the inverse supply/demand (isolate P)

$$4P^D = 1,000 - Q \Rightarrow P^D = 250 - \frac{Q}{4} \text{ (PMB)}$$

$$2P^S = Q + 200 \Rightarrow P^S = \frac{Q}{2} + 100 \text{ (PMC)}$$

$$SMC = PMC + 6 = PMB = SMB$$

$$\frac{Q}{2} + 100 + 6 = 250 - \frac{Q}{4}$$

$$\frac{Q}{2} + \frac{Q}{4} = 250 - 100 - 6$$

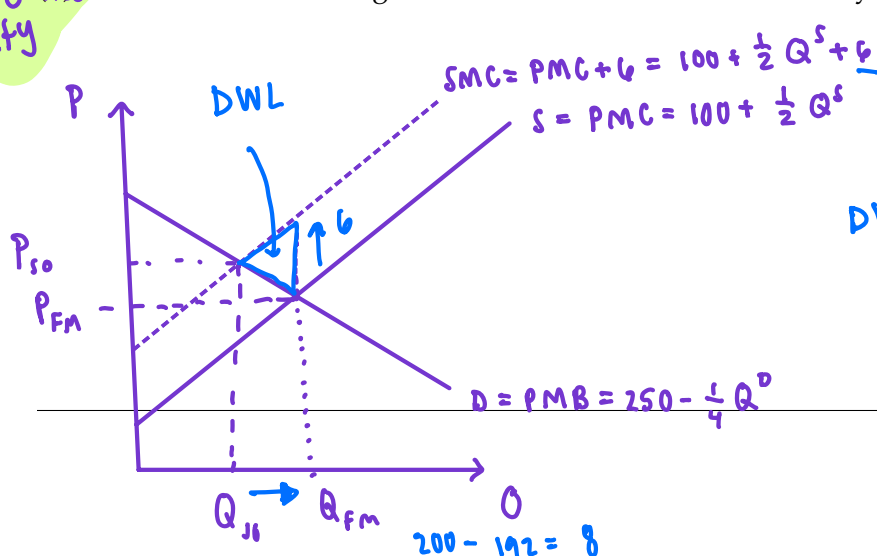
$$\frac{3Q}{4} = 144 \Rightarrow Q_{so} = 144 \cdot \frac{4}{3} = 192$$

if plug into SMC, recall $PMC + 6$

$$\text{or plug into PMB } 250 - \frac{192}{4} = 250 - 48 \Rightarrow P_{so} = 202$$

here, producers internalize the externality

(b) Calculate the deadweight loss associated with the externality.



$$DWL = \frac{1}{2} \cdot 6 \cdot 8 = 24$$

overproduction

1.3 Additional Practice Problems

1.3.1 Based on Gruber Ch.05, Q.14

The marginal damage averted from pollution cleanup is $MD = 200 - 5Q$. The marginal cost associated with pollution cleanup is $MC = 20 + Q$.

→ reduce pollution by one unit and you avoid MD, a benefit!

→ actual cost incurred by the cleaner

(a) What is the optimal level of pollution reduction?

damage averted from pollution cleanup is the benefit of the clean up ($SMB = MD$)

the cost of the cleanup is the PMC and the SMC ($PMC = SMC = MC$)

$$200 - 5Q = 20 + Q \Rightarrow 180 = 6Q \Rightarrow Q^* = 30$$

(b) Show that this level of pollution reduction could be accomplished through taxation. What tax per unit would generate the optimal amount of pollution reduction?

$20 + Q^*$
the MC
of clean up

$$SMC = 20 + 30 = 50$$

to implement social optimum of 30 units of pollution requires a tax of 50 on each unit of pollution

1.3.2 Gruber Ch.05, Q.4

In the midwestern United States, where winds tend to blow from west to east, states tend to more easily approve new polluting industries near their eastern borders than in other parts of the state. Why do you think this is true?